Milestone 2: Initial Descriptive Analysis

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Academic Field

Observations

Academic field is a categorical variable with three levels: Arts, Other, and Prof. Categorical variables are appropriately summarized using counts and proportions, with suitable visualizations being bar charts and mosaic plots, allowing comparison of distributions across groups. The dataset contains 19,792 faculty members: majority fall in the Other category (66.4%), followed by Prof (19.2%) and Arts (14.3%). Across all years, 20.4%, 69.7%, and 9.9% of female faculty were concentrated in Arts, Other, and Prof respectively; for male faculty, the corresponding percentages were 12.8%, 65.6%, and 21.6%. For 1995 only, 19.6%, 70.2%, and 10.3% of female faculty were concentrated in Arts, Other, and Prof respectively; for male faculty, the corresponding percentages were 11.8%, 65.7%, and 22.6%. The distributions are illustrated below using stacked bar charts. A separate bar plot displays the difference in proportions between females and males across all years, with the largest discrepancy occurring in the Prof field.

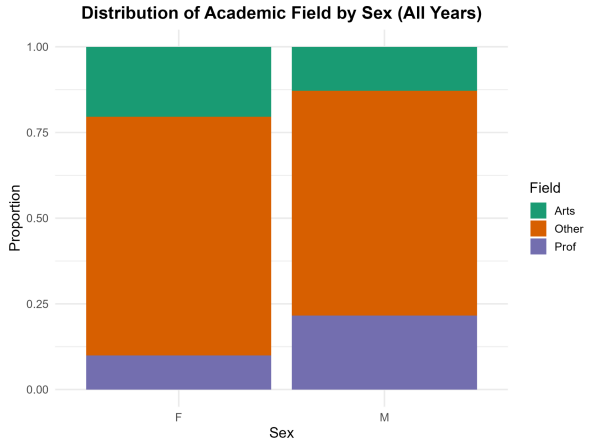

Figure 1: A stacked bar chart describing the distribution of the academic field against sex.

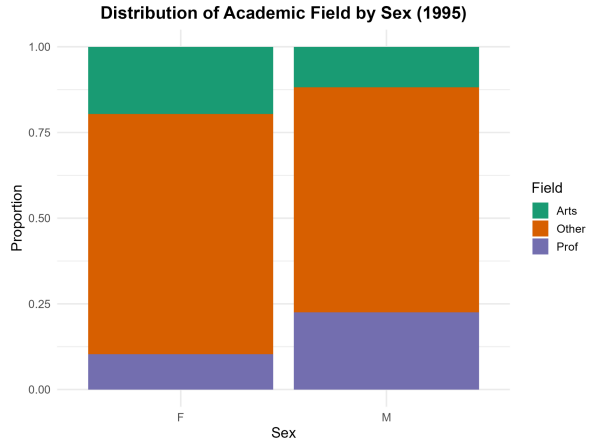

Figure 2: A stacked bar chart describing the distribution of the academic field against sex in 1995.

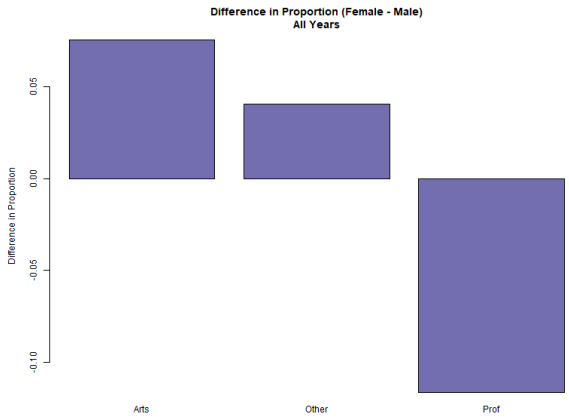

Figure 3: A chart showcasing the difference in proportion across values in the academics field.

Gender-Based Differences in Academic Field

A Chi-square test of independence was conducted to examine whether academic field and sex are associated. The null hypothesis states that academic field and sex are independent, while the alternative states that they are related. For the full dataset, the test produced $\chi^2(2) = 355.79$ with $p < 2.2 * 10^{-16}$; for the 1995 subset, $\chi^2(2) = 37.96$ with $p < 5.73 * 10^{-9}$. In both cases, the p-values are far below the significance level $\alpha = 0.05$, so we reject the null hypothesis and conclude that there is strong statistical evidence of an association between academic field and sex. This implies that men and women are distributed differently across academic fields. Since compensation may differ across academic fields, this uneven distribution may contribute to observed salary differences.

Year Attained Highest Degree

Observations

Year attained highest degree is a quantitative variable, summarized using measures of center and spread. Suitable visualizations include histograms, density plots, and boxplots, which assess skewness, spread, and potential outliers. The overall histogram is left-skewed, meaning most faculty obtained their degrees in later years. When comparing by sex, the overall left-skewness is largely driven by female faculty, whose distribution is more strongly left-skewed and centered on more recent degree years. On average, female faculty obtained their highest degrees later than male faculty (mean: 1981.11 for females versus 1974.37 for males). Numerical and graphical summaries are as below.

Sex	Count	Mean	Median	Standard Deviation
F	409	1981.11	1982	8.7
M	1188	1974.37	1973	9.64

Table 1: Summary Statistics for the Year attained highest degree field

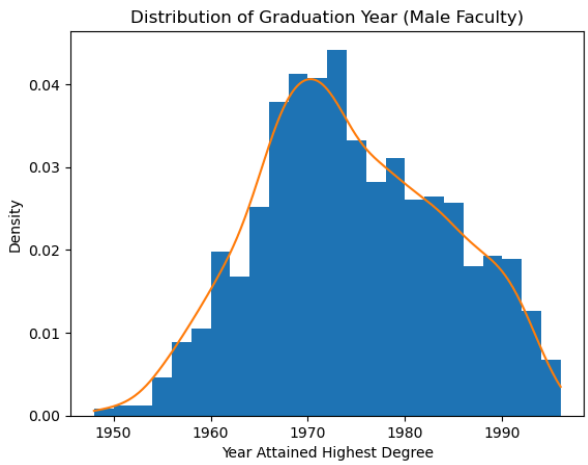

Figure 4 : Showcasing the histogram of the distribution of graduation year for male faculty across years.

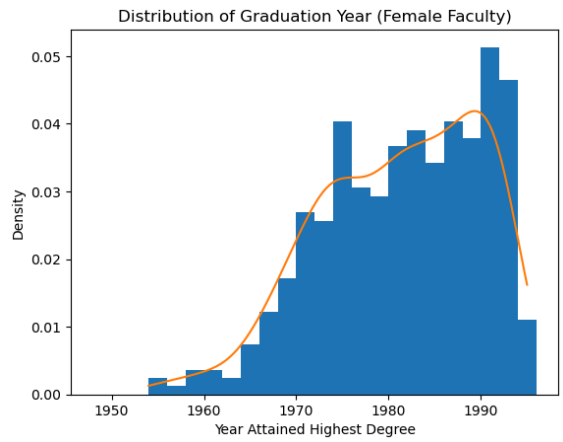

Figure 5: Showcasing the histogram of the distribution of graduation year for female faculty across years.

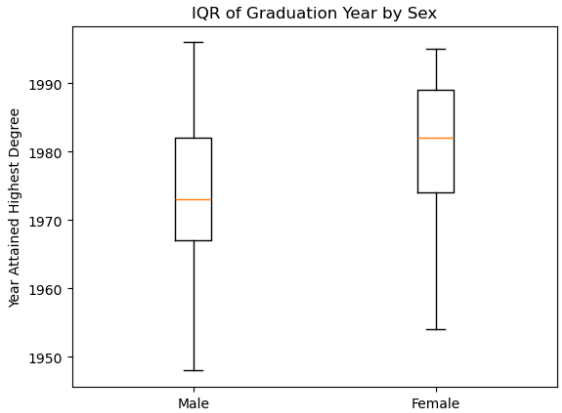

Figure 6: Boxplot showcasing the distribution of the Graduation year by Sex

Gender-Based Differences in Year Attained Highest Degree

To assess whether the average year of highest degree differs between male and female faculty, a Welch two-sample t-test was used, which does not assume equal population variances. Even though the distribution of degree year is not perfectly normal, this test is still appropriate due to the large sample sizes in both groups. The null hypothesis states that the mean degree year is the same for males and females ($\mu_{male} = \mu_{female}$), while the alternative hypothesis states that the means differ ($\mu_{male} \neq \mu_{female}$). The Welch test produced a test statistic of $t =- 13.132$ with a p-value of $1.00 * 10^{-35}$, far below our significance level $\alpha = 0.05$. Therefore, we reject the null hypothesis and conclude that there is extremely strong statistical evidence that the mean degree year differs between male and female faculty. Female faculty, on average, entered academia more recently, reflecting differences in cohort seniority that may influence rank and salary.

Year Hired at the University

Observations

The year hired at the university is a quantitative variable. Likewise, it is summarized using measures of center and spread. Suitable visualizations include histograms, density plots, and boxplots, which assess skewness, spread, and potential outliers. Overall, the distribution of hire year is concentrated in more recent decades with a left-skewed distribution. When comparing by sex, male faculty show a broader spread, including more individuals hired in earlier years. In the side-by-side boxplots, female faculty have a higher median hire year. Across all years, men have been hired more than women and male faculty were hired earlier on average (mean hire 1975 versus 1980.42).

Sex	Count	Mean	Standard Deviation
F	3926	1980.42	8.09
M	15866	1975.00	8.83

Table 2: Summary Statistics for the Year attained highest degree field

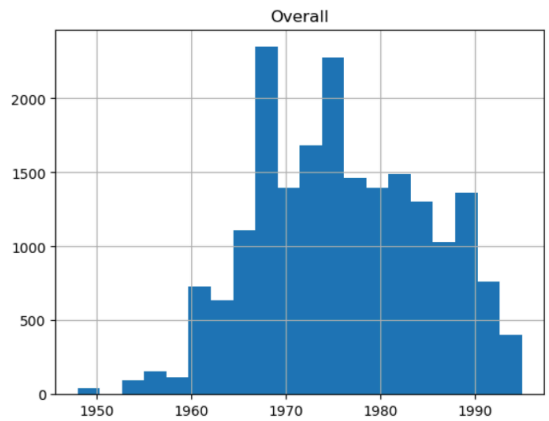

Figure 7: Histogram that describes the year hired at the university across years.

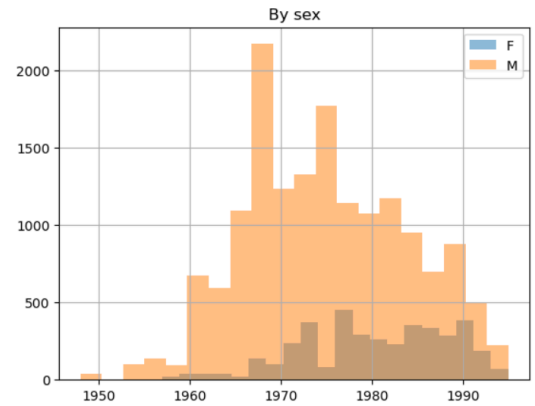

Figure 8: Histogram that describes the year hired at the university across years by Sex.

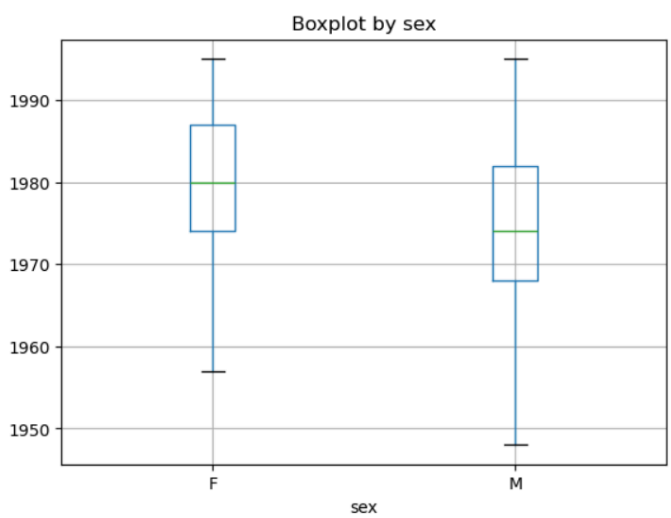

Figure 9: Boxplot that shows the year hired by Sex

Gender-Based Differences in Year Hired at the University

A Welch two-sample t-test was conducted to compare the mean year hired at the university between female and male faculty. The null hypothesis states that the mean hire year is the same for both groups, while the alternative hypothesis states that the means differ, The test produced $t = 36.893$ with p-value $< 2.2 * 10^{-16}$, providing overwhelming evidence against the null hypothesis. Furthermore, the 95% confidence interval for the difference in means $(5.133683,\ 5.709856)$ does not include 0 and indicates that female faculty were hired, on average, about 5 to 6 years later than male faculty.

Academic Rank in 1995

Observations

Academic rank is a categorical variable with three levels: Assistant, Associate, and Full. Categorical variables can be summarized using counts and proportions, and visualized using bar charts. The 1995 data show that most faculty were Full professors. However, the distribution differs by sex: male faculty are more heavily represented at the Full professor level, while female faculty saw a more even distribution across all three levels.

Rank by Sex (Percentage within Sex)			
sex	Assist	Assoc	Full
Female	35.45	33.74	30.81
Male	14.31	25.17	60.52

Table 2: Summary Statistics for the Academic Rank field by Sex

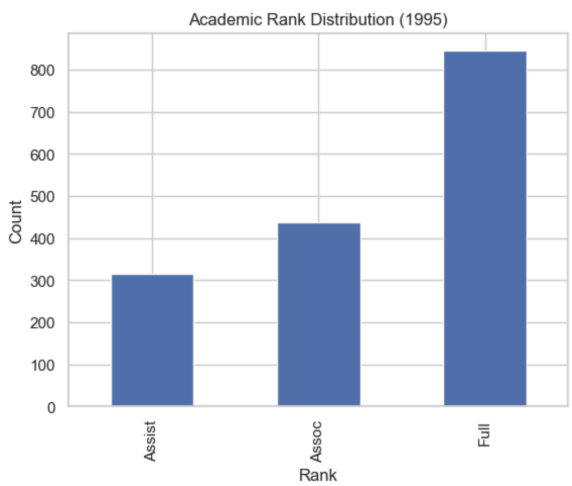

Figure 10: Bar chart showcasing academic rank distribution

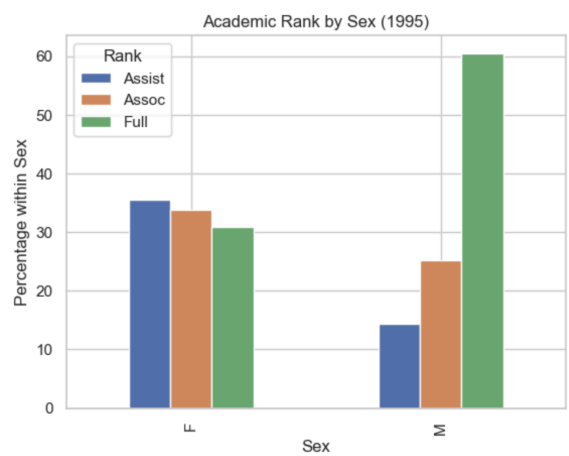

Figure 11: Bar chart for academic rank against sex

Gender-Based Differences in Academic Rank in 1995

To formally assess the relationship between sex and academic rank, a Chi-square test of independence was conducted. The null hypothesis states that academic rank and sex are independent, and the alternative states that academic rank and sex are associated. The test produced a statistic of 127.9 with $p < 2.2 * 10^{-16}$, providing overwhelming evidence against

the null hypothesis. Therefore, we conclude that academic rank and sex are significantly correlated in 1995. Male faculty are more heavily represented at the Full professor level, while female faculty are relatively more concentrated in the Assistant and Associate ranks, suggesting differences in seniority at that time.

Monthly Salary in 1995

Observations

Monthly salary is a quantitative variable. It is summarized using measures of center and spread. Suitable visualizations include histograms, density plots, and boxplots, which assess skewness, spread, and potential outliers. The distribution of monthly salary in 1995 is a t distribution with around 7898.2 degrees of freedom. It is right-skewed, with a smaller number earning substantially higher salaries. Comparing by sex, female faculty earned less on average than male faculty in 1995, with a mean monthly salary of $5,396.91 compared to $6,731.64 for males, and a lower median salary as well.

Sex	Mean (dollars)	Median (dollars)
Female	5396.91	5016
Male	6731.64	6313

Table 3: Summary Statistics for the monthly salary by field Sex

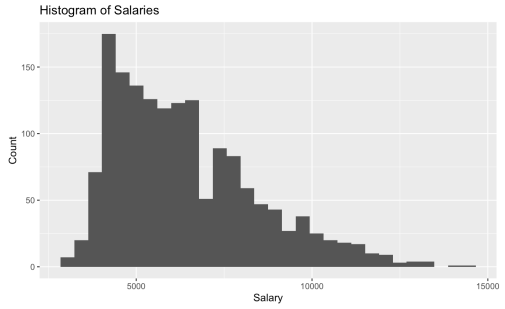

Figure 12: Histogram showcasing distribution of salaries

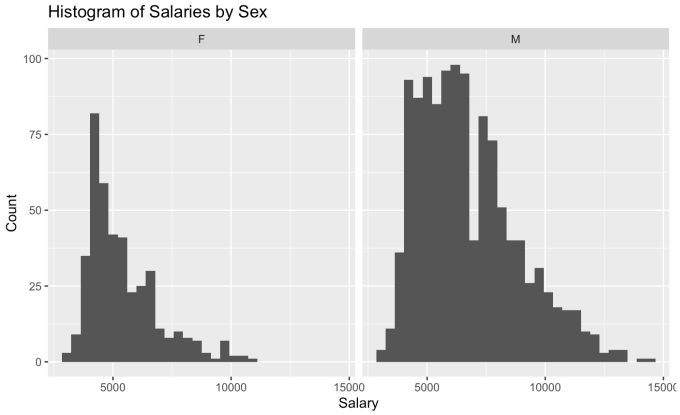

Figure 13: Histogram showcasing distribution of salaries by Sex

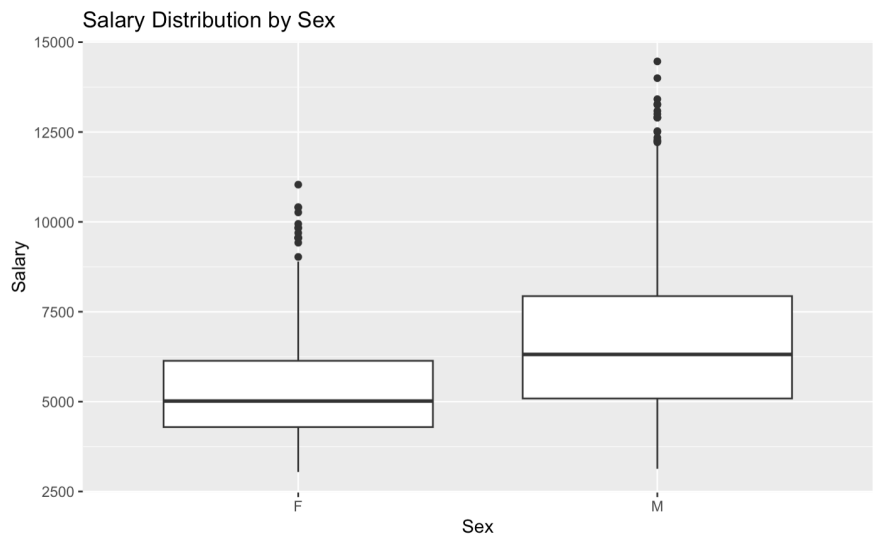

Figure 14: Boxplot that shows the salary distribution by Sex

Gender-Based Differences in Monthly Salary in 1995

To evaluate whether average monthly salary differs between male and female faculty in 1995, a Welch two-sample t-test was conducted, which does not assume equal population variances and uses a student's t-distribution. The null hypothesis states that the mean salary is the same for males and females, while the alternative states that the means differ. The test produced $t =- 14.04$ with a p-value $< 2.2 * 10^{-16}$, providing overwhelming evidence against the null hypothesis. Furthermore, the 95% confidence interval for the difference in means $(- 1521.312, - 1148.151)$ does not contain 0 and indicates that female faculty earned, on average, approximately $1,148 to $1521 less per month than male faculty in 1995.

Conclusion

Overall, the analysis reveals consistent and statistically significant differences between male and female faculty across multiple variables across all years and in 1995. Female faculty were, on average, hired more recently and obtained their highest degrees later than their male counterparts. Women were also more concentrated in lower academic ranks while men were more represented at the Full professor rank and in professional academic fields. A statistically significant gender difference in 1995 monthly salary was observed, with female faculty members earning less on average than male faculty. Our findings from Milestone 2 suggest that differences in field, experience, and rank may contribute to the observed salary gap. Thus, we should consider these systemic factors when interpreting gender differences in faculty compensation.

Code Appendix

Academic Field

```r
library(ggplot2)

salary <- read.table("salary.txt", header = TRUE)

salary$field <- as.factor(salary$field)
salary$sex <- as.factor(salary$sex)

field_palette <- c("#1b9e77", "#d95f02", "#7570b3", "#e7298a", "#66a61e")

base_theme <- theme_minimal(base_size = 14) +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    legend.position = "right"
  )
```

```r
cat("\nACADEMIC FIELD - ALL YEARS\n")
```

```
##
## ACADEMIC FIELD - ALL YEARS
```

```r
field_counts_all <- table(salary$field)
print(field_counts_all)
```

```
##
##  Arts Other  Prof
##  2840 13143  3809
```

```r
field_props_all <- prop.table(field_counts_all)
print(round(field_props_all, 3))
```

```
##
##  Arts Other  Prof
## 0.143 0.664 0.192
```

```r
field_sex_counts_all <- table(salary$field, salary$sex)
print(field_sex_counts_all)
```

```
##
##             F     M
##   Arts    802  2038
##   Other  2735 10408
##   Prof    389  3420
```

```r
field_sex_props_all <- prop.table(field_sex_counts_all, margin = 2)
print(round(field_sex_props_all, 3))
```

```
##
```

```
##             F     M
##   Arts  0.204 0.128
##   Other 0.697 0.656
##   Prof  0.099 0.216
```

```r
print(chisq.test(field_sex_counts_all))
```

```
##
##  Pearson's Chi-squared test
##
## data:  field_sex_counts_all
## X-squared = 355.79, df = 2, p-value < 2.2e-16
```

```r
p1 <- ggplot(salary, aes(x = field)) +
  geom_bar(fill = field_palette[1]) +
  labs(
    title = "Overall Distribution of Academic Field (All Years)",
    x = "Academic Field",
    y = "Count"
  ) +
  base_theme

ggsave("overall_field_counts_all_years.png", p1, width = 8, height = 6)

p2 <- ggplot(salary, aes(x = field)) +
  geom_bar(aes(y = after_stat(count / sum(count))), fill = field_palette[2]) +
  labs(
    title = "Proportion of Faculty by Academic Field (All Years)",
    x = "Academic Field",
    y = "Proportion"
  ) +
  base_theme

ggsave("overall_field_proportion_all_years.png", p2, width = 8, height = 6)

p3 <- ggplot(salary, aes(x = sex, fill = field)) +
  geom_bar(position = "fill") +
  scale_fill_manual(values = field_palette) +
  labs(
    title = "Distribution of Academic Field by Sex (All Years)",
    x = "Sex",
    y = "Proportion",
    fill = "Field"
  ) +
  base_theme

ggsave("stacked_field_by_sex_all_years.png", p3, width = 8, height = 6)
```

```r
data95 <- subset(salary, year == 95)

cat("\nACADEMIC FIELD - 1995 ONLY\n")
```

```
##
## ACADEMIC FIELD - 1995 ONLY
```

```r
field_counts_95 <- table(data95$field)
print(field_counts_95)
```

```
##
##  Arts Other  Prof
##   220  1067   310
```

```r
field_props_95 <- prop.table(field_counts_95)
print(round(field_props_95, 3))
```

```
##
##  Arts Other  Prof
## 0.138 0.668 0.194
```

```r
field_sex_counts_95 <- table(data95$field, data95$sex)
print(field_sex_counts_95)
```

```
##
##           F   M
##   Arts   80 140
##   Other 287 780
##   Prof   42 268
```

```r
field_sex_props_95 <- prop.table(field_sex_counts_95, margin = 2)
print(round(field_sex_props_95, 3))
```

```
##
##             F     M
##   Arts  0.196 0.118
##   Other 0.702 0.657
##   Prof  0.103 0.226
```

```r
print(chisq.test(field_sex_counts_95))
```

```
##
##  Pearson's Chi-squared test
##
## data:  field_sex_counts_95
## X-squared = 37.955, df = 2, p-value = 5.73e-09
```

```r
p4 <- ggplot(data95, aes(x = field)) +
  geom_bar(fill = field_palette[1]) +
  labs(
    title = "Overall Distribution of Academic Field (1995)",
    x = "Academic Field",
    y = "Count"
  ) +
  base_theme

ggsave("overall_field_counts_1995.png", p4, width = 8, height = 6)

p5 <- ggplot(data95, aes(x = sex, fill = field)) +
  geom_bar(position = "fill") +
  scale_fill_manual(values = field_palette) +
  labs(
    title = "Distribution of Academic Field by Sex (1995)",
    x = "Sex",
```

```r
    y = "Proportion",
    fill = "Field"
  ) +
  base_theme

ggsave("stacked_field_by_sex_1995.png", p5, width = 8, height = 6)

png("mosaic_field_by_sex_all_years.png", width = 800, height = 600)
mosaicplot(field ~ sex, data = salary,
           main = "Mosaic Plot of Academic Field by Sex (All Years)",
           col = field_palette)
dev.off()
```

```
## pdf
##   2
```

```r
png("mosaic_field_by_sex_1995.png", width = 800, height = 600)
mosaicplot(field ~ sex, data = data95,
           main = "Mosaic Plot of Academic Field by Sex (1995)",
           col = field_palette)
dev.off()
```

```
## pdf
##   2
```

```r
png("difference_in_proportions_all_years.png", width = 800, height = 600)

diff_props <- field_sex_props_all[, "F"] - field_sex_props_all[, "M"]

barplot(diff_props,
        main = "Difference in Proportion (Female - Male)\nAll Years",
        ylab = "Difference in Proportion",
        col = field_palette[3])

dev.off()
```

```
## pdf
##   2
```

Year Attained Highest Degree

```r
library(readr)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(ggplot2)

col_names_11 <- c(
  "case","id","sex","deg","yrdeg","field",
  "startyr","year","rank","admin","salary"
)

df <- read.table(
  "salary.txt",
  header = TRUE,
  col.names = col_names_11,
  fill = TRUE,
  strip.white = TRUE,
  stringsAsFactors = FALSE
)

str(df)
```

```
## 'data.frame':    19792 obs. of  11 variables:
##  $ case   : int  1 2 3 4 5 6 7 8 9 10 ...
##  $ id     : int  1 2 2 4 6 6 6 6 6 7 ...
##  $ sex    : chr  "F" "M" "M" "M" ...
##  $ deg    : chr  "Other" "Other" "Other" "PhD" ...
##  $ yrdeg  : int  92 91 91 96 66 66 66 66 66 70 ...
##  $ field  : chr  "Other" "Other" "Other" "Other" ...
##  $ startyr: int  95 94 94 95 91 91 91 91 91 71 ...
##  $ year   : int  95 94 95 95 91 92 93 94 95 76 ...
##  $ rank   : chr  "Assist" "Assist" "Assist" "Assist" ...
##  $ admin  : int  0 0 0 0 1 1 0 0 0 0 ...
##  $ salary : num  6684 4743 4881 4231 11182 ...
```

```r
head(df)
```

```
##   case id sex   deg yrdeg field startyr year   rank admin salary
## 1    1  1   F Other    92 Other      95   95 Assist     0   6684
## 2    2  2   M Other    91 Other      94   94 Assist     0   4743
## 3    3  2   M Other    91 Other      94   95 Assist     0   4881
## 4    4  4   M   PhD    96 Other      95   95 Assist     0   4231
## 5    5  6   M   PhD    66 Other      91   91   Full     1  11182
## 6    6  6   M   PhD    66 Other      91   92   Full     1  11507
```

```r
print(min(df$yrdeg, na.rm = TRUE))
```

```
## [1] 48
```

```r
print(max(df$yrdeg, na.rm = TRUE))
```

```
## [1] 96
```

```r
df <- df %>%
  mutate(yrdeg = 1900 + yrdeg)

head(df)
```

```
##   case id sex   deg yrdeg field startyr year   rank admin salary
## 1    1  1   F Other  1992 Other      95   95 Assist     0   6684
## 2    2  2   M Other  1991 Other      94   94 Assist     0   4743
```

```
## 3    3  2   M Other  1991 Other      94   95 Assist     0   4881
## 4    4  4   M   PhD  1996 Other      95   95 Assist     0   4231
## 5    5  6   M   PhD  1966 Other      91   91   Full     1  11182
## 6    6  6   M   PhD  1966 Other      91   92   Full     1  11507
```

```
df1 <- df %>%
  arrange(id, year) %>%
  group_by(id) %>%
  summarise(
    sex   = first(sex),
    yrdeg = first(yrdeg),
    .groups = "drop"
  )

head(df1)
```

```
## # A tibble: 6 x 3
##      id sex   yrdeg
##   <int> <chr> <dbl>
## 1     1 F      1992
## 2     2 M      1991
## 3     4 M      1996
## 4     6 M      1966
## 5     7 M      1970
## 6     8 M      1975
```

```
colSums(is.na(df1))
```

```
##    id   sex yrdeg
##     0     0     0
```

```
male   <- df1 %>% filter(sex == "M") %>% pull(yrdeg)
female <- df1 %>% filter(sex == "F") %>% pull(yrdeg)

cat("Male n:", length(male), "\n")
```

```
## Male n: 1188
```

```
cat("Female n:", length(female), "\n")
```

```
## Female n: 409
```

```
male_mean   <- mean(male)
female_mean <- mean(female)

male_median   <- median(male)
female_median <- median(female)

cat("Male mean yrdeg:", male_mean, "\n")
```

```
## Male mean yrdeg: 1974.369
```

```
cat("Female mean yrdeg:", female_mean, "\n")
```

```
## Female mean yrdeg: 1981.108
```

```
cat("Difference in means (M - F):", male_mean - female_mean, "\n")
```

```
## Difference in means (M - F): -6.738893
```

```r
bins <- seq(min(df1$yrdeg), max(df1$yrdeg) + 2, by = 2)

ggplot(df1, aes(x = yrdeg)) +
  geom_histogram(breaks = bins) +
  labs(
    x = "Year Attained Highest Degree",
    y = "Number of Faculty",
    title = "Distribution of Graduation Year (All Faculty)"
  )
```

```r
ggplot(data.frame(yrdeg = male), aes(x = yrdeg)) +
  geom_histogram(aes(y = after_stat(density)), breaks = bins) +
  geom_density() +
  labs(
    x = "Year Attained Highest Degree",
    y = "Density",
    title = "Distribution of Graduation Year (Male Faculty)"
  )
```

```r
ggplot(data.frame(yrdeg = female), aes(x = yrdeg)) +
  geom_histogram(aes(y = after_stat(density)), breaks = bins) +
  geom_density() +
  labs(
    x = "Year Attained Highest Degree",
    y = "Density",
    title = "Distribution of Graduation Year (Female Faculty)"
  )
```

```r
df1 %>%
  group_by(sex) %>%
  summarise(
    count  = n(),
    mean   = mean(yrdeg),
    median = median(yrdeg),
    std    = sd(yrdeg),
    .groups = "drop"
  )
```

```
## # A tibble: 2 x 5
##   sex   count  mean median   std
##   <chr> <int> <dbl>  <dbl> <dbl>
## 1 F       409 1981.   1982  8.70
## 2 M      1188 1974.   1973  9.64
```

```r
boxplot(
  male, female,
  names = c("Male", "Female"),
  ylab = "Year Attained Highest Degree",
  main = "IQR of Graduation Year by Sex"
)
```

```r
tt <- t.test(male, female, var.equal = FALSE)

cat("t-statistic:", unname(tt$statistic), "\n")
```

```
## t-statistic: -13.13177
```

```r
cat("p-value:", tt$p.value, "\n")
```

```
## p-value: 1.002651e-35
```

Year Hired at the University

```r
df <- read.table("salary.txt", header = TRUE)

head(df)
```

```
##   case id sex   deg yrdeg field startyr year   rank admin salary
## 1    1  1   F Other    92 Other      95   95 Assist     0   6684
## 2    2  2   M Other    91 Other      94   94 Assist     0   4743
## 3    3  2   M Other    91 Other      94   95 Assist     0   4881
## 4    4  4   M   PhD    96 Other      95   95 Assist     0   4231
## 5    5  6   M   PhD    66 Other      91   91   Full     1  11182
## 6    6  6   M   PhD    66 Other      91   92   Full     1  11507
```

```r
df$year_hired <- 1900 + df$startyr

head(df[, c("sex", "startyr", "year_hired")])
```

```
##   sex startyr year_hired
## 1   F      95       1995
## 2   M      94       1994
## 3   M      94       1994
## 4   M      95       1995
## 5   M      91       1991
## 6   M      91       1991
```

```r
df %>%
  group_by(sex) %>%
  summarise(
    count = n(),
    mean  = mean(year_hired, na.rm = TRUE),
    std   = sd(year_hired, na.rm = TRUE)
  )
```

```
## # A tibble: 2 x 4
##   sex   count  mean   std
##   <chr> <int> <dbl> <dbl>
## 1 F      3926 1980.  8.09
## 2 M     15866 1975.  8.83
```

```r
c(count = length(df$year_hired), mean = mean(df$year_hired,
                                             na.rm = TRUE),
  std = sd(df$year_hired, na.rm = TRUE))
```

```
##        count         mean          std
## 19792.000000  1976.072201     8.951341
```

```r
cat("Overall\n")
```

```
## Overall
```

```r
print(summary(df$year_hired))
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1948    1969    1976    1976    1983    1995
```

```r
cat("\nBy sex\n")
```

```
##
## By sex
```

```r
print(by(df$year_hired, df$sex, summary))
```

```
## df$sex: F
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1957    1974    1980    1980    1987    1995
## ------------------------------------------------------------
## df$sex: M
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1948    1968    1974    1975    1982    1995
```

```r
rng <- range(df$year_hired, na.rm = TRUE)
breaks <- seq(rng[1], rng[2], length.out = 21)

blue   <- "#1f77b4"
orange <- "#ff7f0e"

add_mpl_grid <- function() {
  usr <- par("usr")
  x_ticks <- axTicks(1)
  y_ticks <- axTicks(2)
  abline(v = x_ticks, col = "grey70", lwd = 1)
  abline(h = y_ticks, col = "grey70", lwd = 1)
}

hist(df$year_hired,
     breaks = breaks,
     right = FALSE,
     col = blue,
     border = NA,
     main = "Overall",
     xlab = "",
     ylab = "")
add_mpl_grid()
```

```r
hF <- hist(df$year_hired[df$sex == "F"], breaks = breaks, right = FALSE, plot = FALSE)
hM <- hist(df$year_hired[df$sex == "M"], breaks = breaks, right = FALSE, plot = FALSE)
ylim_shared <- c(0, max(c(hF$counts, hM$counts)))

breaksF <- seq(min(df$year_hired[df$sex=="F"]), max(df$year_hired[df$sex=="F"]), length.out=21)
breaksM <- seq(min(df$year_hired[df$sex=="M"]), max(df$year_hired[df$sex=="M"]), length.out=21)

hF <- hist(df$year_hired[df$sex=="F"], breaks=breaksF, right=FALSE, plot=FALSE)
hM <- hist(df$year_hired[df$sex=="M"], breaks=breaksM, right=FALSE, plot=FALSE)
ylim_shared <- c(0, max(c(hF$counts, hM$counts)))

hist(df$year_hired[df$sex=="F"], breaks=breaksF, right=FALSE,
     col=adjustcolor("#1f77b4", 0.5), border=NA, ylim=ylim_shared,
     xlim=c(1950, max(df$year_hired, na.rm=TRUE)),
```

```r
     main="By sex", xlab="", ylab="")

hist(df$year_hired[df$sex=="M"], breaks=breaksM, right=FALSE,
     col=adjustcolor("#ff7f0e", 0.5), border=NA, add=TRUE)

legend("topright", legend=c("F","M"),
       fill=c(adjustcolor("#1f77b4",0.5), adjustcolor("#ff7f0e",0.5)),
       bty="n")
add_mpl_grid()
```

```r
boxplot(year_hired ~ sex,
        data = df,
        main = "Boxplot by sex",
        xlab = "",
        ylab = "")
```

```r
df$sex <- as.factor(df$sex)

welch_hire <- t.test(year_hired ~ sex, data = df, var.equal = FALSE)

welch_hire
```

```
##
##  Welch Two Sample t-test
##
## data:  year_hired by sex
## t = 36.893, df = 6437.4, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group F and group M is not equal to 0
## 95 percent confidence interval:
##  5.133683 5.709856
## sample estimates:
## mean in group F mean in group M
##        1980.418        1974.997
```

Academic Rank in 1995

```r
df <- read.table("salary.txt", header = TRUE)
head(df, 5)
```

```
##   case id sex   deg yrdeg field startyr year   rank admin salary
## 1    1  1   F Other    92 Other      95   95 Assist     0   6684
## 2    2  2   M Other    91 Other      94   94 Assist     0   4743
## 3    3  2   M Other    91 Other      94   95 Assist     0   4881
## 4    4  4   M   PhD    96 Other      95   95 Assist     0   4231
## 5    5  6   M   PhD    66 Other      91   91   Full     1  11182
```

```r
df_95 <- df %>% filter(year == 95)

cat("Number of faculty in 1995:", nrow(df_95), "\n")
```

```
## Number of faculty in 1995: 1597
```

```r
rank_counts <- sort(table(df_95$rank))
rank_percent <- prop.table(table(df_95$rank)) * 100
rank_percent <- rank_percent[names(rank_counts)]
```

```r
rank_summary <- data.frame(
  Count = as.integer(rank_counts),
  Percentage = round(as.numeric(rank_percent), 2),
  row.names = names(rank_counts)
)

cat("\nAcademic Rank Distribution (1995):\n")
```

```
##
## Academic Rank Distribution (1995):
```

```r
print(rank_summary)
```

```
##        Count Percentage
## Assist   315      19.72
## Assoc    437      27.36
## Full     845      52.91
```

```r
rank_sex_table <- table(df_95$rank, df_95$sex)
rank_sex_percent <- prop.table(rank_sex_table, margin = 2) * 100

cat("\nRank by Sex (Counts):\n")
```

```
##
## Rank by Sex (Counts):
```

```r
print(rank_sex_table)
```

```
##
##            F   M
##   Assist 145 170
##   Assoc  138 299
##   Full   126 719
```

```r
cat("\nRank by Sex (Percentage within Sex):\n")
```

```
##
## Rank by Sex (Percentage within Sex):
```

```r
print(round(rank_sex_percent, 2))
```

```
##
##              F     M
##   Assist 35.45 14.31
##   Assoc  33.74 25.17
##   Full   30.81 60.52
```

```r
barplot(rank_counts,
        main = "Academic Rank Distribution (1995)",
        xlab = "Rank",
        ylab = "Count")
```

```r
barplot(t(rank_sex_percent),
        beside = TRUE,
        legend.text = TRUE,
        main = "Academic Rank by Sex (Percentage within Sex)",
        xlab = "Sex",
        ylab = "Percentage")
```

```r
field_counts <- sort(table(df_95$field), decreasing = TRUE)
field_percent <- prop.table(table(df_95$field)) * 100
field_percent <- field_percent[names(field_counts)]

cat("\nAcademic Field Distribution (1995):\n")
```

```
##
## Academic Field Distribution (1995):
```

```r
print(data.frame(
  Count = as.integer(field_counts),
  Percentage = round(as.numeric(field_percent), 2),
  row.names = names(field_counts)
))
```

```
##       Count Percentage
## Other  1067      66.81
## Prof    310      19.41
## Arts    220      13.78
```

```r
prop.table(table(df_95$field, df_95$sex), margin = 2) * 100
```

```
##
##                F        M
##   Arts  19.55990 11.78451
##   Other 70.17115 65.65657
##   Prof  10.26895 22.55892
```

```r
plt_field_counts <- sort(table(df_95$field), decreasing = TRUE)
barplot(plt_field_counts,
        main = "Academic Field Distribution (1995)",
        xlab = "Field",
        ylab = "Count")
```

```r
cat("\nYear Highest Degree Summary:\n")
```

```
##
## Year Highest Degree Summary:
```

```r
print(summary(df_95$yrdeg))
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   48.00   69.00   76.00   76.09   84.00   96.00
```

```r
breaks_yrdeg <- seq(min(df_95$yrdeg, na.rm = TRUE),
                    max(df_95$yrdeg, na.rm = TRUE),
                    length.out = 16)

hist(df_95$yrdeg,
     breaks = breaks_yrdeg,
     col = "#4C72B0",
     border = "white",
     main = "Distribution of Year Highest Degree Attained",
     xlab = "Year",
     ylab = "Frequency")
grid()
```

```r
cat("\nStart Year Summary:\n")
```

```
##
## Start Year Summary:
```

```r
print(summary(df_95$startyr))
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   48.00   73.00   83.00   81.12   90.00   95.00
```

```r
breaks_startyr <- seq(min(df_95$startyr, na.rm = TRUE),
                      max(df_95$startyr, na.rm = TRUE),
                      length.out = 16)

hist(df_95$startyr,
     breaks = breaks_startyr,
     col = "#4C72B0",
     border = "white",
     main = "Distribution of Year Hired",
     xlab = "Start Year",
     ylab = "Frequency")
grid()
```

```r
cat("\nSalary Summary (1995):\n")
```

```
##
## Salary Summary (1995):
```

```r
print(summary(df_95$salary))
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    3042    4743    5962    6390    7602   14464
```

```r
breaks_salary <- seq(min(df_95$salary, na.rm = TRUE),
                     max(df_95$salary, na.rm = TRUE),
                     length.out = 21)

hist(df_95$salary,
     breaks = breaks_salary,
     col = "#4C72B0",
     border = "white",
     main = "Salary Distribution (1995)",
     xlab = "Monthly Salary",
     ylab = "Frequency")
grid()
```

```r
cat("\nSalary by Sex:\n")
```

```
##
## Salary by Sex:
```

```r
print(df_95 %>% group_by(sex) %>% summarise(
  count = n(),
  mean = mean(salary, na.rm = TRUE),
  sd = sd(salary, na.rm = TRUE),
  min = min(salary, na.rm = TRUE),
  `25%` = quantile(salary, 0.25, na.rm = TRUE),
```

```r
  `50%` = quantile(salary, 0.50, na.rm = TRUE),
  `75%` = quantile(salary, 0.75, na.rm = TRUE),
  max = max(salary, na.rm = TRUE)
))
```

```
## # A tibble: 2 x 9
##   sex   count  mean    sd   min `25%` `50%` `75%`   max
##   <chr> <int> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 F       409 5397. 1481. 3042   4292  5016  6135 11036
## 2 M      1188 6732. 2090. 3131.  5088  6313  7935 14464
```

```r
boxplot(salary ~ sex, data = df_95,
        main = "Salary by Sex (1995)",
        xlab = "Sex",
        ylab = "Salary")
```

```r
df_95 <- subset(df, year == 95)

chisq.test(table(df_95$rank, df_95$sex))
```

```
##
##  Pearson's Chi-squared test
##
## data:  table(df_95$rank, df_95$sex)
## X-squared = 127.9, df = 2, p-value < 2.2e-16
```

Monthly Salary in 1995

```r
df <- read.table("salary.txt", header = TRUE, sep = "")
df <- df %>% filter(year == 95)
head(df)
```

```
##   case id sex   deg yrdeg field startyr year   rank admin    salary
## 1    1  1   F Other    92 Other      95   95 Assist     0  6684.000
## 2    3  2   M Other    91 Other      94   95 Assist     0  4881.000
## 3    4  4   M   PhD    96 Other      95   95 Assist     0  4231.000
## 4    9  6   M   PhD    66 Other      91   95   Full     0 12184.000
## 5   29  7   M   PhD    70 Other      71   95  Assoc     0  4604.000
## 6   30  8   M   PhD    75 Other      95   95 Assist     0  4047.674
```

```r
missing_counts_by_group <- df %>%
  group_by(sex) %>%
  summarise(
    Missing_Value = sum(is.na(salary))
    )

# Print the result
print(missing_counts_by_group)
```

```
## # A tibble: 2 x 2
##   sex   Missing_Value
##   <chr>         <int>
## 1 F                 0
## 2 M                 0
```

```r
mean_salary <- mean(df$salary, na.rm = TRUE)
median_salary <- median(df$salary, na.rm = TRUE)
mean_salary
```

```
## [1] 6389.808
```

```r
median_salary
```

```
## [1] 5962
```

```r
df %>%
  group_by(sex) %>%
  summarise(
    mean_salary = mean(salary, na.rm = TRUE),
    median_salary = median(salary, na.rm = TRUE)
    )
```

```
## # A tibble: 2 x 3
##   sex   mean_salary median_salary
##   <chr>       <dbl>         <dbl>
## 1 F           5397.          5016
## 2 M           6732.          6313
```

```r
ggplot(df, aes(x = salary)) +
  geom_histogram(bins = 30) +
  theme_gray() +
  labs(
    title = "Histogram of Salaries",
    x = "Salary",
    y = "Count"
    )
```

```r
ggplot(df, aes(x = salary)) +
  geom_histogram(bins = 30) +
  facet_wrap(~ sex) +
  theme_gray()+
  labs(
    title = "Histogram of Salaries by Sex",
    x = "Salary",
    y = "Count"
    )
```

```r
ggplot(df, aes(x = sex, y = salary)) +
  geom_boxplot() +
  theme_grey() +
  labs(
    title = "Salary Distribution by Sex",
    x = "Sex",
    y = "Salary"
    )
```

```r
df_95 <- subset(df, year == 95)

welch_salary <- t.test(salary ~ sex, data = df_95, var.equal = FALSE)

welch_salary
```

```
##
##  Welch Two Sample t-test
##
## data:  salary by sex
## t = -14.038, df = 997.73, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group F and group M is not equal to 0
## 95 percent confidence interval:
##  -1521.312 -1148.151
## sample estimates:
## mean in group F mean in group M
##        5396.908        6731.640
```